Chapter 5

Count Regression

When the response is an unbounded count (0,1,2,3,...), we can use a count regression model to explain this in terms of the given predictors. Sometimes other models may be appropriate. When the count is bounded, a binomial-type response regression as discussed in the previous chapters is sensible. In some cases, the counts might be sufficiently large that a normal approximation is justified so that a normal linear model may be used. We shall consider two distributions for counts: the Poisson and, less commonly, the negative binomial.

5.1 Poisson Regression

If Y is Poisson with mean $\mu > 0$, then:

$$P(Y = y) = \frac{e^{-\mu}\mu^y}{y!}, \qquad y = 0, 1, 2, \ldots$$

Three examples of the Poisson density are depicted in Figure 5.1. In the left panel, we see a distribution that gives highest probability to $y = 0$ and falls rapidly as y increases. In the center panel, we see a skew distribution with longer tail on the right. Even for a not so large $\mu = 5$, we see the distribution become more normally shaped. This becomes more pronounced as μ increases.

```
barplot(dpois(0:5,0.5),xlab="y",ylab="Probability",names=0:5,main="
    ↪ mean = 0.5")
barplot(dpois(0:10,2),xlab="y",ylab="Probability",names=0:10,main="
    ↪ mean = 2")
barplot(dpois(0:15,5),xlab="y",ylab="Probability",names=0:15,main="
    ↪ mean = 5")
```

The expectation and variance of a Poisson are the same: $EY = \text{var}\, Y = \mu$. The Poisson distribution arises naturally in several ways:

1. If the count is some number out of some possible total, then the response would be more appropriately modeled as a binomial. However, for small success probabilities and large totals, the Poisson is a good approximation and can be used. For example, in modeling the incidence of rare forms of cancer, the number of people affected is a small proportion of the population in a given geographical area. Specifically, if $\mu = np$ while $n \to \infty$, then $B(n, p)$ is well approximated by $Pois(\mu)$. Also, for small p, note that $\text{logit}(p) \approx \log p$, so that the use of the Poisson with a log link is comparable to the binomial with a logit link. Where n varies between cases, a *rate model* can be used as described in Section 5.3.

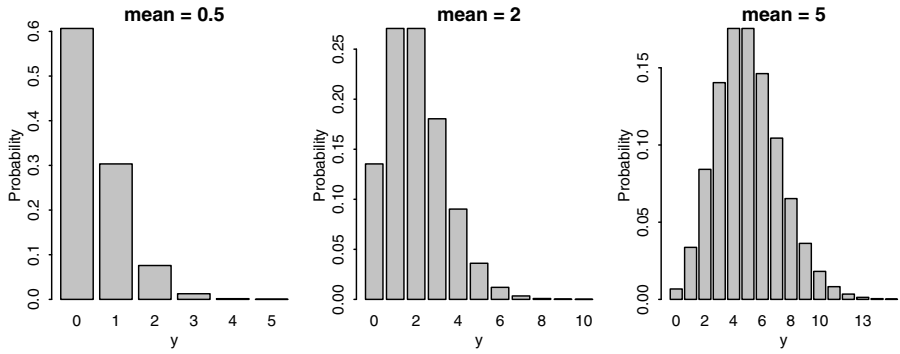


Figure 5.1 *Poisson probabilities for* $\mu = 0.5$*,* 2 *and* 5*, respectively.*

2. Suppose the probability of occurrence of an event in a given time interval is proportional to the length of that time interval and independent of the occurrence of other events. In this case the number of events in any specified time interval will be Poisson distributed. Examples include modeling the number of incoming telephone calls to a service center or the number of earthquakes in a fixed period of time. However, in any real application, the assumptions are likely to be violated. For example, the rate of incoming telephone calls is likely to vary with the time of day while the timing of earthquakes are unlikely to be completely independent. Nevertheless, the Poisson may be a good approximation.
3. Poisson distributions also arise naturally when the time between events is independent and identically exponentially distributed. We count the number of events in a given time period. This is effectively equivalent to the previous case, since the exponential distribution between events will result from the assumption of constant and independent probability of occurrence of an event in an interval.

If the count is the number falling into some level of a given category, then a multinomial response model or categorical data analysis should be used. For example, if we have counts of how many people have type O, A, B or AB blood and are interested in how that relates to race and gender, then a straight Poisson regression model will not be appropriate. We will see later that the Poisson distribution still comes into play in Chapter 7.

An important result concerning Poisson random variables is that their sum is also Poisson. Specifically, suppose that $Y_i \sim Pois(\mu_i)$ for $i = 1, 2, \ldots$ and are independent, then $\sum_i Y_i \sim Pois(\sum_i \mu_i)$. This is useful because sometimes we have access only to the aggregated data. If we assume the individual-level data is Poisson, then so is the summed data and Poisson regression can still be applied.

For 30 Galápagos Islands, we have a count of the number of plant species found on each island and the number that are endemic to that island. We also have five geographic variables for each island. The data was presented by Johnson and Raven (1973) and also appear in Weisberg (2005). We have filled in a few missing values

that appeared in the original dataset for simplicity. We model the number of species using normal linear regression:

```
data(gala, package="faraway")
gala <- gala[,-2]
```

We throw out the Endemics variable (which falls in the second column of the dataframe) since we won't be using it in this analysis. We fit a linear regression and look at the residual vs. fitted plot:

```
modl <- lm(Species ~ . , gala)
plot(modl, 1)
```

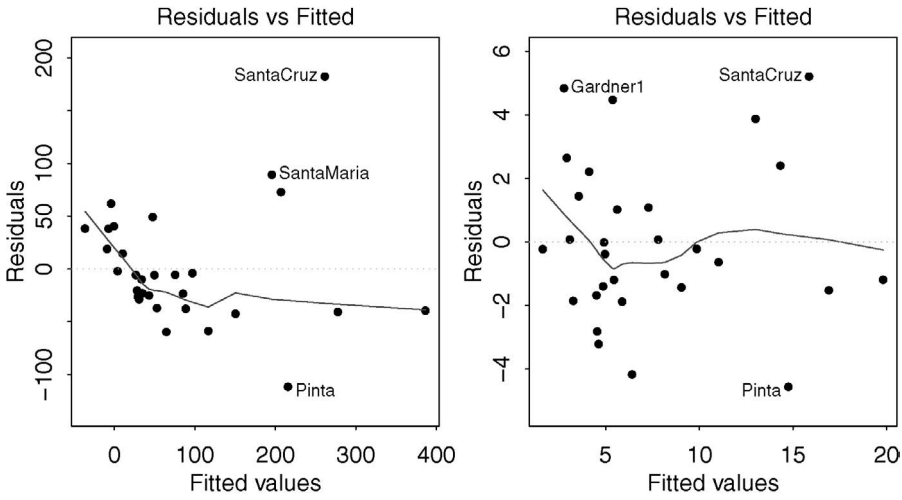


Figure 5.2 *Residual-fitted plots for the Galápagos dataset. The plot on the left is for a model with the original response while that on the right is for the square-root transformed response.*

We see clear evidence of nonconstant variance in the left panel of Figure 5.2. Some experimentation (or the use of the Box–Cox method) reveals that a square-root transformation is best:

```
modt <- lm(sqrt(Species) ~ . , gala)
plot(modt, 1)
```

We now see in the right panel of Figure 5.2 that the nonconstant variance problem has been cleared up. Let's take a look at the fit:

```
library(faraway)
sumary(modt)
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.391924   0.871268    3.89  0.00069
Area        -0.001972   0.001020   -1.93  0.06508
Elevation    0.016478   0.002441    6.75  5.5e-07
Nearest      0.024933   0.047950    0.52  0.60784
Scruz       -0.013483   0.009798   -1.38  0.18151
Adjacent    -0.003367   0.000805   -4.18  0.00033

n = 30, p = 6, Residual SE = 2.774, R-Squared = 0.78
```

We see a fairly good fit ($R^2 = 0.78$) considering the nature of the variables. However, we achieved this fit at the cost of transforming the response. This makes interpretation more difficult. Furthermore, some of the response values are quite small (single digits) which makes us question the validity of the normal approximation. This model may be adequate, but perhaps we can do better. We develop a Poisson regression model.

Suppose we have count responses Y_i that we wish to model in terms of a vector of predictors x_i. Now if $Y_i \sim Pois(\mu_i)$, we need some way to link the μ_i to the x_i. We use a linear combination of the x_i to form the linear predictor $\eta_i = x_i^T\beta$. Since we require that $\mu_i \geq 0$, we can ensure this by using a log link function, that is:

$$\log\mu_i = \eta_i = x_i^T\beta$$

So, as with the binomial regression models of the previous chapter, this model also has a linear predictor and a link function. The log-likelihood is:

$$l(\beta) = \sum_{i=1}^{n}(y_i x_i^T\beta - \exp(x_i^T\beta) - \log(y_i!))$$

Differentiating with respect to β_j gives the MLE as the solution to:

$$\sum_{i=1}^{n}(y_i - \exp(x_i^T\hat{\beta}))x_{ij} = 0 \qquad \forall j$$

which can be more compactly written as:

$$X^T y = X^T\hat{\mu}$$

The normal equations for the least squares estimate of β in Gaussian linear models take the same form when we set $\hat{\mu} = X\hat{\beta}$. The equations for β for a binomial regression with a logit link also take the same form. This would not be true for other link functions. The link function having this property is known as the *canonical link*.

However, there is no explicit formula for $\hat{\beta}$ for the Poisson (or binomial) regression and we must resort to numerical methods to find a solution. We fit the Poisson regression model to the Galápagos data:

```
modp <- glm(Species ~ ., family=poisson, gala)
sumary(modp)
```

```
              Estimate Std. Error z value  Pr(>|z|)
(Intercept)  3.1548079  0.0517495   60.96   < 2e-16
Area        -0.0005799  0.0000263  -22.07   < 2e-16
Elevation    0.0035406  0.0000874   40.51   < 2e-16
Nearest      0.0088256  0.0018213    4.85 0.0000013
Scruz       -0.0057094  0.0006256   -9.13   < 2e-16
Adjacent    -0.0006630  0.0000293  -22.61   < 2e-16

n = 30 p = 6
Deviance = 716.846 Null Deviance = 3510.729 (Difference = 2793.883)
```

Using the same arguments as for binary response regression as seen in Section 2.3, we develop a deviance for the Poisson regression:

$$D = 2\sum_{i=1}^{n}(y_i\log(y_i/\hat{\mu}_i) - (y_i - \hat{\mu}_i))$$

This Poisson deviance is also known as the *G-statistic*.

The same asymptotic inference may be employed as for the binomial model. We can judge the goodness of fit of a proposed model by checking the deviance of the model against a χ^2 distribution with degrees of freedom equal to that of the model. We can compare nested models by taking the difference of the deviances and comparing to a χ^2 distribution with degrees of freedom equal to the difference in the number of parameters for the two models. We can test the significance of individual predictors and construct confidence intervals for β using the standard errors, $se(\hat{\beta})$, although, as before, it is better to use profile likelihood methods.

An alternative and perhaps better-known goodness of fit measure is the Pearson's X^2 statistic:

$$X^2 = \sum_{i=1}^{n} \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i}$$

In this example, we see that the residual deviance is 717 on 24 degrees of freedom, which indicates an ill-fitting model if indeed the Poisson is the correct model for the response. A common drawback in using the standard Poisson regression is that responses are often more variable than the Poisson model would imply. The Poisson distribution has only one parameter μ which represents both the mean and the variance. In contrast, the standard linear model has a variance σ^2 that is independent of the mean μ and hence is more flexible.

5.2 Dispersed Poisson Model

We can modify the standard Poisson model to allow for more variation in the response. But before we do that, we must check whether the large size of deviance might be related to some other cause.

In the Galápagos example, we check the residuals to see if the large deviance can be explained by an outlier:

```
halfnorm(residuals(modp))
```

The half-normal plot of the (absolute value of the) residuals shown in Figure 5.3 shows no outliers. It could be that the structural form of the model needs some improvement, but some experimentation with different forms for the predictors will reveal that there is little scope for improvement. Furthermore, the proportion of deviance explained by this model, $1 - 717/3510 = 0.796$, is about the same as in the linear model above.

For a Poisson distribution, the mean is equal to the variance. Let's investigate this relationship for this model. It is difficult to estimate the variance for a given value of the mean, but $(y - \hat{\mu})^2$ does serve as a crude approximation. We plot this estimated variance against the mean, as seen in the second panel of Figure 5.3:

```
plot(log(fitted(modp)),log((gala$Species-fitted(modp))^2), xlab=
    ↪ expression(hat(mu)),ylab=expression((y-hat(mu))^2))
abline(0,1)
```

We see that the variance is proportional to, but larger than, the mean. When the variance assumption of the Poisson regression model is broken but the link function and choice of predictors are correct, the estimates of β are consistent, but the standard er-

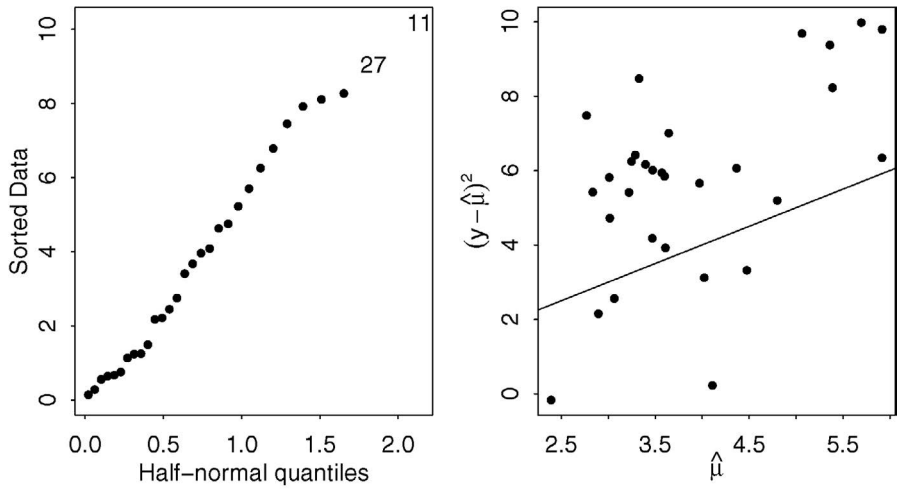


Figure 5.3 *Half-normal plot of the residuals of the Poisson model is shown on the left and the relationship between the mean and variance is shown on the right. A line representing mean equal to variance is also shown.*

rors will be wrong. We cannot determine which predictors are statistically significant in the above model using the output we have.

The Poisson distribution has only one parameter and so is not very flexible for empirical fitting purposes. We can generalize by allowing ourselves a dispersion parameter. Over- or underdispersion can occur in various ways in Poisson models. For example, suppose the Poisson response Y has rate λ which is itself a random variable. The tendency to fail for a machine may vary from unit to unit even though they are the same model. We can model this by letting λ be gamma distributed with $E\lambda = \mu$ and var $\lambda = \mu/\phi$. Now Y is negative binomial with mean $EY = \mu$. The mean is the same as the Poisson, but the variance var $Y = \mu(1+\phi)/\phi$ which is not equal to μ. In this case, overdispersion would occur and could be modeled using a negative binomial model as demonstrated in Section 5.4.

If we know the specific mechanism, as in the above example, we could model the response as a negative binomial or other more flexible distribution. If the mechanism is not known, we can introduce a dispersion parameter ϕ such that var $Y = \phi EY = \phi\mu$. $\phi = 1$ is the regular Poisson regression case, while $\phi > 1$ is overdispersion and $\phi < 1$ is underdispersion.

The dispersion parameter may be estimated using:

$$\hat{\phi} = \frac{X^2}{n-p} = \frac{\sum_i (y_i - \hat{\mu}_i)^2/\hat{\mu}_i}{n-p}$$

We estimate the dispersion parameter in our example by:

```
(dp <- sum(residuals(modp,type="pearson")^2)/modp$df.res)
[1] 31.749
```

We can then adjust the standard errors and so forth in the summary as follows:

```
sumary(modp,dispersion=dp)
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.154808   0.291590   10.82  < 2e-16
Area        -0.000580   0.000148   -3.92  8.9e-05
Elevation    0.003541   0.000493    7.19  6.5e-13
Nearest      0.008826   0.010262    0.86     0.39
Scruz       -0.005709   0.003525   -1.62     0.11
Adjacent    -0.000663   0.000165   -4.01  6.0e-05

overdispersion parameter = 31.749
n = 30 p = 6
Deviance = 716.846 Null Deviance = 3510.729 (Difference = 2793.883)
```

Notice that the estimation of the dispersion and the regression parameters is independent, so choosing a dispersion other than one has no effect on the regression parameter estimates. Notice also that there is some similarity in which variables are picked out as significant when compared with the linear regression model.

The same result can be achieved by using the *quasi-Poisson* model which directly incorporates the dispersion parameter:

```
modd <- glm(Species ~ ., family=quasipoisson, gala)
```

When comparing Poisson models with overdispersion, an F-test rather than a χ^2 test should be used. As in normal linear models, the variance, or dispersion parameter in this case, needs to be estimated. This requires the use of the F-test. We can test the significance of each of the predictors relative to the full model:

```
drop1(modd,test="F")
Single term deletions

Model:
Species ~ Area + Elevation + Nearest + Scruz + Adjacent
          Df Deviance  AIC F value  Pr(>F)
<none>             717  890
Area       1     1204 1375   16.32 0.00048
Elevation  1     2390 2560   56.00   1e-07
Nearest    1      739  910    0.76 0.39336
Scruz      1      814  984    3.24 0.08444
Adjacent   1     1341 1512   20.91 0.00012
```

The z-statistics from the `sumary()` are less reliable and so the F-test is preferred. In this example, there is little practical difference between the two.

5.3 Rate Models

The number of events observed may depend on a size variable that determines the number of opportunities for the events to occur. For example, if we record the number of burglaries reported in different cities, the observed number will depend on the number of households in these cities. In other cases, the size variable may be time. For example, if we record the number of customers served by a sales worker, we must take account of the differing amounts of time worked.

Sometimes, it is possible to analyze such data using a binomial response model. For the burglary example above, we might model the number of burglaries out of the number of households. However, if the proportion is small, the Poisson approxima-

tion to the binomial is effective. Furthermore, in some examples, the total number of potential cases may not be known exactly. The modeling of rare diseases illustrates this issue as we may know the number of cases but not have precise population data. Sometimes, the binomial model simply cannot be used. In the burglary example, some households may be robbed more than once. In the customer service example, the size variable is not a count. An alternative approach is to model the ratio. However, there are often difficulties with normality and unequal variance when taking this approach, particularly if the counts are small.

In Purott and Reeder (1976), some data is presented from an experiment conducted to determine the effect of gamma radiation on the numbers of chromosomal abnormalities (`ca`) observed. The number (`cells`), in hundreds of cells exposed in each run, differs. The dose amount (`doseamt`) and the rate (`doserate`) at which the dose is applied are the predictors of interest. We may format the data for observation like this:

```
data(dicentric, package="faraway")
round(xtabs(ca/cells ~ doseamt+doserate, dicentric),2)
```

```
       doserate
doseamt  0.1 0.25  0.5    1  1.5    2  2.5    3    4
    1   0.05 0.05 0.07 0.07 0.06 0.07 0.07 0.07 0.07
    2.5 0.16 0.28 0.29 0.32 0.38 0.41 0.41 0.37 0.44
    5   0.48 0.82 0.90 0.88 1.23 1.32 1.34 1.24 1.43
```

We can plot the data as seen in the first panel of Figure 5.4:

```
with(dicentric,interaction.plot(doseamt, doserate, ca/cells, legend=
    ↪ FALSE))
```

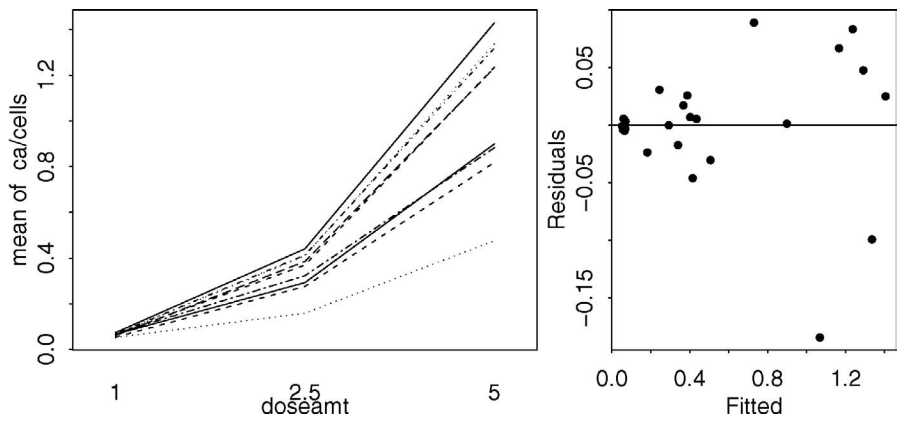


Figure 5.4 *Chromosomal abnormalities rate response is shown on the left and a residuals vs. fitted plot of a linear model fit to these data is shown on the right.*

We might try modeling the rate directly. We see that the effect of the dose rate may be multiplicative, so we log this variable in the following model:

```
lmod <- lm(ca/cells ~ log(doserate)*factor(doseamt), dicentric)
summary(lmod)$adj
```

```
[1] 0.98444
```

As can be seen from the adjusted R^2, this model fits well. However, a look at the diagnostics reveals a problem, as seen in the second panel of Figure 5.4:

```
plot(residuals(lmod) ~ fitted(lmod),xlab="Fitted",ylab="Residuals")
abline(h=0)
```

We might prefer an approach that directly models the count response. We need to use the log of the number of cells because we expect this to have a multiplicative effect on the response:

```
dicentric$dosef <- factor(dicentric$doseamt)
pmod <- glm(ca ~ log(cells)+log(doserate)*dosef,  family=poisson,
    ↪ dicentric)
sumary(pmod)
```

```
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)           -2.7653     0.3812   -7.25    4e-13
log(cells)             1.0025     0.0514   19.52  < 2e-16
log(doserate)          0.0720     0.0355    2.03  0.04240
dosef2.5               1.6298     0.1027   15.87  < 2e-16
dosef5                 2.7667     0.1229   22.52  < 2e-16
log(doserate):dosef2.5 0.1611     0.0484    3.33  0.00087
log(doserate):dosef5   0.1932     0.0430    4.49    7e-06

n = 27 p = 7
Deviance = 21.748 Null Deviance = 916.127 (Difference = 894.379)
```

We can relate this Poisson model with a log link back to a linear model for the ratio response:

$$\log(\texttt{ca}/\texttt{cells}) = X\beta$$

This can be rearranged as

$$\log \texttt{ca} = \log \texttt{cells} + X\beta$$

We are using log `cells` as a predictor. Checking above, we can see that the coefficient of 1.0025 is very close to one. This suggests fitting a model with the coefficient fixed as one. In this manner, we are modeling the rate of chromosomal abnormalities while still maintaining the count response for the Poisson model. This is known as a *rate model*. We fix the coefficient as one by using an *offset*. Such a term on the predictor side of the model equation has no parameter attached:

```
rmod <- glm(ca ~ offset(log(cells))+log(doserate)*dosef,  family=
    ↪ poisson,dicentric)
sumary(rmod)
```

```
                     Estimate Std. Error z value  Pr(>|z|)
(Intercept)           -2.7467     0.0343  -80.16   < 2e-16
log(doserate)          0.0718     0.0352    2.04   0.04130
dosef2.5               1.6254     0.0495   32.86   < 2e-16
dosef5                 2.7611     0.0435   63.49   < 2e-16
log(doserate):dosef2.5 0.1612     0.0483    3.34   0.00084
log(doserate):dosef5   0.1935     0.0424    4.56 0.0000051

n = 27 p = 6
Deviance = 21.750 Null Deviance = 4753.004 (Difference = 4731.254)
```

Not surprisingly, the coefficients are only slightly different from the previous model. We see from the residual deviance that the model fits well. Previous analyses have posited a quadratic effect in dose; indeed, the observed coefficients speak against

a purely linear effect. However, given that we have only three dose levels, we can hardly check whether quadratic is appropriate. Given the significant interaction effect, we can see that the effect of the dose rate is different depending on the overall dose. We can see that the combination of a high dose, delivered quickly, has a greater combined effect than the main effect estimates would suggest. More on the analysis of this data may be found in Frome and DuFrain (1986).

5.4 Negative Binomial

Given a series of independent trials, each with probability of success p, let Z be the number of trials until the kth success. Then:

$$P(Z=z) = \binom{z-1}{k-1} p^k (1-p)^{z-k} \qquad z = k, k+1, \ldots$$

The negative binomial can arise naturally in several ways. Imagine a system that can withstand k hits before failing. The probability of a hit in a given time period is p and we count the number of time periods until failure. The negative binomial also arises from a generalization of the Poisson where the parameter λ is gamma distributed. The negative binomial also comes up as a limiting distribution for urn schemes that can be used to model contagion.

We get a more convenient parameterization if we let $Y = Z - k$ and $p = (1+\alpha)^{-1}$ so that:

$$P(Y=y) = \binom{y+k-1}{k-1} \frac{\alpha^y}{(1+\alpha)^{y+k}}, \qquad y = 0, 1, 2, \ldots$$

then $EY = \mu = k\alpha$ and $\operatorname{var} Y = k\alpha + k\alpha^2 = \mu + \mu^2/k$.

The log-likelihood is then:

$$\sum_{i=1}^{n} \left(y_i \log \frac{\alpha}{1+\alpha} - k \log(1+\alpha) + \sum_{j=0}^{y_i - 1} \log(j+k) - \log(y_i!) \right)$$

The most convenient way to link the mean response μ to a linear combination of the predictors X is:

$$\eta = x^T \beta = \log \frac{\alpha}{1+\alpha} = \log \frac{\mu}{\mu + k}$$

We can regard k as fixed and determined by the application or as an additional parameter to be estimated. More on regression models for negative binomial responses may be found in Cameron and Trivedi (1998) and Lawless (1987).

Consider this example. ATT ran an experiment varying five factors relevant to a wave-soldering procedure for mounting components on printed circuit boards. The response variable, `skips`, is a count of how many solder skips appeared to a visual inspection. The data comes from Comizzoli et al. (1990). We start with a Poisson regression:

```
data(solder, package="faraway")
modp <- glm(skips ~ . , family=poisson, data=solder)
c(deviance(modp), df.residual(modp))
```

```
[1] 1829  882
```

We see that the full model has a residual deviance of 1829 on 882 degrees of freedom. This is not a good fit. Perhaps including interaction terms will improve the fit:

```
modp2  <- glm(skips ~ (Opening +Solder + Mask + PadType + Panel)^2 ,
    ↪ family=poisson, data=solder)
deviance(modp2)
```

```
[1] 1068.8
```

```
pchisq(deviance(modp2),df.residual(modp2),lower=FALSE)
```

```
[1] 1.1307e-10
```

The fit is improved but not enough to conclude that the model fits. We could try adding more interactions but that would make interpretation increasingly difficult. A check for outliers reveals no problem.

An alternative model for counts is the negative binomial. The functions for fitting come from the `MASS` package — see Venables and Ripley (2002) for more details. We can specify the link parameter k. Here we choose $k = 1$ to demonstrate the method, although there is no substantive motivation from this application to use this value. Note that the $k = 1$ case corresponds to an assumption of a *geometric* distribution for the response.

```
library(MASS)
modn <- glm(skips ~ .,negative.binomial(1),solder)
modn
```

```
Coefficients:
(Intercept)     OpeningM     OpeningS   SolderThin       MaskA3
    -1.6993       0.5085       1.9997       1.0489       0.6571
     MaskA6       MaskB3       MaskB6    PadTypeD6    PadTypeD7
     2.5265       1.2726       2.0803      -0.4612       0.0161
  PadTypeL4    PadTypeL6    PadTypeL7    PadTypeL8    PadTypeL9
     0.4688      -0.4711      -0.2949      -0.0849      -0.5213
  PadTypeW4    PadTypeW9        Panel
    -0.1425      -1.4836       0.1693

Degrees of Freedom: 899 Total (i.e. Null);  882 Residual
Null Deviance:      1740
Residual Deviance: 559  AIC: 3880
```

We could experiment with different values of k, but there is a more direct way of achieving this by allowing the parameter k to vary and be estimated using maximum likelihood in:

```
modn <- glm.nb(skips ~ .,solder)
summary(modn)
```

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.4225     0.1427   -9.97  < 2e-16
OpeningM      0.5029     0.0798    6.31  2.9e-10
OpeningS      1.9132     0.0715   26.75  < 2e-16
SolderThin    0.9393     0.0536   17.52  < 2e-16
MaskA3        0.5898     0.0965    6.11  9.9e-10
MaskA6        2.2673     0.1018   22.27  < 2e-16
MaskB3        1.2110     0.0964   12.57  < 2e-16
MaskB6        1.9904     0.0922   21.58  < 2e-16
PadTypeD6    -0.4659     0.1124   -4.15  3.4e-05
PadTypeD7    -0.0331     0.1067   -0.31  0.75611
PadTypeL4     0.3827     0.1026    3.73  0.00019
PadTypeL6    -0.5784     0.1141   -5.07  4.0e-07
```

```
PadTypeL7    -0.3666     0.1109   -3.30  0.00095
PadTypeL8    -0.1589     0.1082   -1.47  0.14199
PadTypeL9    -0.5660     0.1139   -4.97  6.8e-07
PadTypeW4    -0.2004     0.1087   -1.84  0.06526
PadTypeW9    -1.5646     0.1362  -11.49  < 2e-16
Panel         0.1637     0.0314    5.21  1.8e-07

(Dispersion parameter for Negative Binomial(4.3972) family taken to be 1)

    Null deviance: 4043.3  on 899  degrees of freedom
Residual deviance: 1008.3  on 882  degrees of freedom
AIC: 3683
             Theta:  4.397
         Std. Err.:  0.495
 2 x log-likelihood:  -3645.309
```

We see that $\hat{k} = 4.397$ with a standard error of 0.495. We can compare negative binomial models using the usual inferential techniques.

5.5 Zero Inflated Count Models

Sometimes we see count response data where the number of zeroes appearing is significantly greater than the Poisson or negative binomial models would predict. Consider the number of arrests for criminal offenses incurred by individuals. A large number of people have never been arrested by the police while a smaller number have been detained on multiple occasions. Modifying the Poisson by adding a dispersion parameter does not adequately model this divergence from the standard count distributions.

We consider a sample of 915 biochemistry graduate students as analyzed by Long (1990). The response is the number of articles produced during the last three years of the PhD. We are interested in how this is related to the gender, marital status, number of children, prestige of the department and productivity of the advisor of the student. The dataset may be found in the `pscl` package of Zeileis et al. (2008) which also provides the new model fitting functions needed in this section. We start by fitting a Poisson regression model:

```
library(pscl)
modp <- glm(art ~ ., data=bioChemists, family=poisson)
sumary(modp)
```

```
            Estimate Std. Error z value  Pr(>|z|)
(Intercept)  0.30462    0.10298    2.96    0.0031
femWomen    -0.22459    0.05461   -4.11 0.0000392
marMarried   0.15524    0.06137    2.53    0.0114
kid5        -0.18488    0.04013   -4.61 0.0000041
phd          0.01282    0.02640    0.49    0.6271
ment         0.02554    0.00201   12.73   < 2e-16

n = 915 p = 6
Deviance = 1634.371 Null Deviance = 1817.405 (Difference = 183.034)
```

We can see that deviance is significantly larger than the degrees of freedom. Some experimentation reveals that this cannot be solved by using a richer linear predictor or by eliminating some outliers. We might consider a dispersed Poisson model or negative binomial but some thought suggests that there are good reasons why a

student might produce no articles at all. We count and predict how many students produce between zero and seven articles. Very few students produce more than seven articles so we ignore these. The `predprob` function produces the predicted probabilities for each case. By summing these, we get the expected number for each article count.

```
ocount <- table(bioChemists$art)[1:8]
pcount <- colSums(predprob(modp)[,1:8])
plot(pcount,ocount,type="n",xlab="Predicted",ylab="Observed")
text(pcount,ocount, 0:7)
```

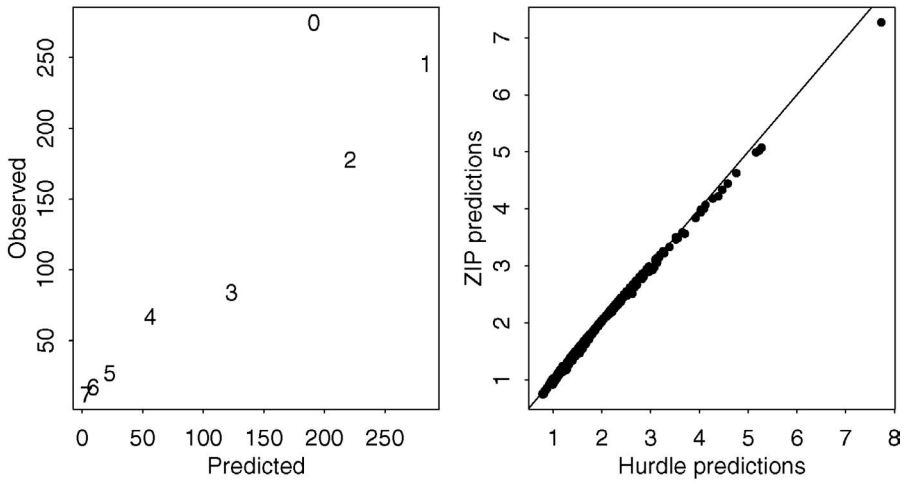


Figure 5.5 *On the left, the predicted and observed counts for number of articles 0-7 is shown. On the right, the predictions from the hurdle and zero-inflated Poisson model are compared.*

In the left panel of Figure 5.5, we see that there are many more unproductive students than would be predicted by the Poisson model. In contrast, the relationship between observed and predicted is linear for the students who produce at least one article.

There are two main ways of modeling an excess of zero counts. First we consider the *hurdle* model. This can be understood in terms of a latent variable. In our example, this latent variable might measure the propensity of the student to publish. We can imagine a hurdle which this latent variable must exceed for a publication to be produced. If the latent variable is less than the hurdle, nothing appears, but as it exceeds the hurdle, more articles may be produced. The latent variable is related to the predictors in some manner we may wish to uncover. A general specification of the model is:

$$P(Y=0) = f_1(0)$$
$$P(Y=j) = \frac{1-f_1(0)}{1-f_2(0)} f_2(j), \qquad j>0$$

The first part of the model describes the probability of zero being observed (or not). We shall use a binary response model that can link this probability to the predictors.

The second part of the model specifies the probability of outcomes greater than zero. We shall use a Poisson for f_2 but this now describes a *truncated* Poisson, as zero is not an allowable outcome and we must rescale the distribution accordingly. Zero is the hurdle here but we could specify higher hurdles if the application warranted it. We can fit such a model as:

```
modh <- hurdle(art ~ ., data=bioChemists)
summary(modh)
```

```
Count model coefficients (truncated poisson with log link):
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.67114    0.12246    5.48  4.2e-08
femWomen    -0.22858    0.06522   -3.51  0.00046
marMarried   0.09648    0.07283    1.32  0.18521
kid5        -0.14219    0.04845   -2.93  0.00334
phd         -0.01273    0.03130   -0.41  0.68434
ment         0.01875    0.00228    8.22  < 2e-16
Zero hurdle model coefficients (binomial with logit link):
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.2368     0.2955    0.80    0.423
femWomen     -0.2512     0.1591   -1.58    0.114
marMarried    0.3262     0.1808    1.80    0.071
kid5         -0.2852     0.1111   -2.57    0.010
phd           0.0222     0.0796    0.28    0.780
ment          0.0801     0.0130    6.15  7.5e-10

Number of iterations in BFGS optimization: 12
Log-likelihood: -1.61e+03 on 12 Df
```

Notice that the model produces two sets of coefficients — one for the truncated Poisson part of the model and one for the binary response part of the model. As can be seen, these can be quite different. By default a binomial model with a logit link is used for the zero component and a Poisson with a log link is used for the count component. Sensible alternatives can be specified. Before interpreting the model, we present a second form of model for zero inflated count data.

Suppose we ask people how many games of chess they have played in the last month. Some will say zero because they do not play chess but some zero responses will be from chess players who have not managed a game in the last month. Circumstances such as these require a *mixture* model. A general specification of this model takes the form:

$$P(Y=0) = \phi + (1-\phi)f(0)$$
$$P(Y=j) = (1-\phi)f(j), \qquad j>0$$

The parameter ϕ represents the proportion who will always respond zero. We can model this proportion using binary response model. The distribution f models the counts of those individuals that can have a positive response. In some cases, these individuals will have a zero response which combines with the always-zero subjects. We can use a Poisson for f in which case this is called *zero-inflated Poisson* or ZIP model. We can fit this:

```
modz <- zeroinfl(art ~ ., data=bioChemists)
summary(modz)
```

```
Count model coefficients (poisson with log link):
            Estimate Std. Error z value Pr(>|z|)
```

```
(Intercept)  0.64084    0.12131    5.28  1.3e-07
femWomen    -0.20914    0.06340   -3.30  0.00097
marMarried   0.10375    0.07111    1.46  0.14456
kid5        -0.14332    0.04743   -3.02  0.00251
phd         -0.00617    0.03101   -0.20  0.84238
ment         0.01810    0.00229    7.89  3.1e-15

Zero-inflation model coefficients (binomial with logit link):
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.57706    0.50939   -1.13    0.257
femWomen     0.10975    0.28008    0.39    0.695
marMarried  -0.35401    0.31761   -1.11    0.265
kid5         0.21710    0.19648    1.10    0.269
phd          0.00127    0.14526    0.01    0.993
ment        -0.13411    0.04524   -2.96    0.003

Number of iterations in BFGS optimization: 21
Log-likelihood: -1.6e+03 on 12 Df
```

The results take a similar form to the hurdle model. Focusing on the zero part of the model, we notice that the `ment` variable which counts the number of articles produced by the mentor is the most significant predictor in both cases but takes opposing signs. This is because the hurdle approach models the probability of a positive count whereas the zero-inflated approach models the probability of a zero count. Hence there is no contradiction.

It is difficult to choose between the two approaches. We can compare the fitted values:

```
plot(fitted(modh), fitted(modz), xlab="Hurdle predictions", ylab="ZIP
    ↪ predictions")
abline(0,1)
```

As can be seen in the second panel of Figure 5.5, there is not much difference between these fits. We might use our background knowledge of the application to make an appropriate choice.

We can use the standard likelihood testing theory to compare nested models. For example, suppose we consider a simplified version of the ZIP model where we now have different predictors for the two components of the model. The count part of the model is specified before the | and the zero model after.

```
modz2 <- zeroinfl(art ~ fem+kid5+ment | ment, data=bioChemists)
summary(modz2)
Count model coefficients (poisson with log link):
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.69452    0.05303   13.10  < 2e-16
femWomen    -0.23386    0.05840   -4.00  6.2e-05
kid5        -0.12652    0.03967   -3.19   0.0014
ment         0.01800    0.00222    8.10  5.7e-16

Zero-inflation model coefficients (binomial with logit link):
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.6849     0.2053   -3.34  0.00085
ment         -0.1268     0.0398   -3.18  0.00145

Number of iterations in BFGS optimization: 13
Log-likelihood: -1.61e+03 on 6 Df
```

Twice the difference in the log-likelihood will be approximately χ^2-distributed with degrees of freedom equal to the difference in the number of parameters.

```
(lrt <- 2*(modz$loglik-modz2$loglik))
[1] 6.1728
1-pchisq(6.1728,6)
[1] 0.40411
```

Given the large p-value of 0.4, we conclude that our simplification of the model is justifiable. For interpretation, the exponentiated coefficients are more useful:

```
exp(coef(modz2))
count_(Intercept)    count_femWomen        count_kid5
          2.00274           0.79147           0.88116
       count_ment  zero_(Intercept)         zero_ment
          1.01817           0.50415           0.88091
```

We see that among students who are inclined to write articles, women produce at a rate 0.79 times that for men and each child under five reduces production by about 12% while each additional article produced by the mentor increases production by about 1.8%. We see that each extra article from the mentor reduces the odds of a nonproductive student by a factor of 0.88. Of course, none of these interpretations can be taken as causal.

We can also use the model to make predictions. Consider a single male with no children whose mentor produced six articles:

```
newman <- data.frame(fem="Men",mar="Single",kid5=0,ment=6)
predict(modz2, newdata=newman, type="prob")
        0       1       2       3        4        5        6
1 0.27759 0.19394 0.21636 0.16091 0.089758 0.040054 0.014895
...
```

We see that most likely outcome for this student is that no articles will be produced with a probability of 0.278. We can query the probability of no production from the zero part of the model:

```
predict(modz2, newdata=newman, type="zero")
      1
0.19067
```

So the additional probability to make this up to 0.278 comes from the Poisson count part of the model. This difference might be attributed to the student who had the potential to write an article but just didn't do it.

Exercises

1. The dataset `discoveries` lists the numbers of "great" inventions and scientific discoveries in each year from 1860 to 1959.
   (a) Plot the discoveries over time and comment on the trend, if any.
   (b) Fit a Poisson response model with a constant term. Now compute the mean number of discoveries per year. What is the relationship between this mean and the coefficient seen in the model?
   (c) Use the deviance from the model to check whether the model fits the data. What does this say about whether the rate of discoveries is constant over time?
   (d) Make a table of how many years had zero, one, two, three, etc. discoveries. Collapse eight or more into a single category. Under an appropriate Poisson

distribution, calculate the expected number of years with each number of discoveries. Plot the observed against the expected using a different plotting character to denote the number of discoveries. How well do they agree?

(e) Use the Pearson's Chi-squared test to check whether the observed numbers are consistent with the expected numbers. Interpret the result.

(f) Fit a Poisson response model that is quadratic in the year. Test for the significance of the quadratic term. What does this say about the presence of a trend in discovery?

(g) Compute the predicted number of discoveries each year and show these predictions as a line drawn over the data. Comment on what you see.

2. The `salmonella` data was collected in a salmonella reverse mutagenicity assay. The predictor is the dose level of quinoline and the response is the numbers of revertant colonies of TA98 salmonella observed on each of three replicate plates.

(a) Plot the data and comment on the relationship between dose and colonies.

(b) Compute the mean and variance within each set of observations with the same dose. Plot the variance against the mean and comment on what this says about overdispersion.

(c) Fit a model with dose treated as a six-level factor. Check the deviance to determine whether this model fits the data. Do you think it is possible to find a transformation of the dose predictor that results in a Poisson model that does fit the data?

(d) Make a QQ plot of the residuals from the previous model. Interpret the plot.

(e) Fit a Poisson model what includes an overdispersion parameter and is quadratic in the dose. Can we determine from the deviance of this model whether the fit is adequate?

(f) Plot the residuals against the fitted values for the previous model. Interpret the plot.

(g) Plot the fitted mean response of this model on top of the data.

(h) Give the predicted mean response for a dose of 500. Compute a 95% confidence interval.

(i) At what dose does the maximum predicted response occur?

3. The `ships` dataset found in the `MASS` package gives the number of wave damage incidents and aggregate months of service for different types of ships broken down by year of construction and period of operation.

(a) Examine the data for the period of operation 1960–1974 for ships constructed in the years 1975–1979. Why are there no damage incidents?

(b) Make a two-way table that shows the rate of damage incidents per 1000 months of aggregate service classified by type and year of construction. Comment on the table.

(c) Compute the rate of damage incidents per year for all cases where some service was recorded.

(d) Fit linear models with the observed rate of damage incidents as the response and the following three combinations of predictors: (i) All two-way interactions, (ii) main effects terms only, (iii) null (no predictors). Make sure year is treated as a factor rather than numerical variable. Which of these three models is preferred?

(e) Fit a Poisson response model for the number of incidents with the predictors: log of service, type, year and period. Test whether the parameter associated with the service term can be one. Explain why we are interested in such a test.

(f) Fit the Poisson rate model with all two-way interactions of the three predictors. Does this model fit the data?

(g) Check the residuals. Are there any outliers? Plot residuals against the fitted values. Investigate any unusual features of this plot.

(h) Now fit the rate model with just the main effects and compare it to the interaction model. Which model is preferred?

(i) Fit quasi-Poisson versions of the two previous models and repeat the comparison.

(j) Interpret the coefficients of the main effects of the quasi-Poisson model. What factors are associated with higher and lower rates of damage incidents?

4. The dataset `africa` gives information about the number of military coups in sub-Saharan Africa and various political and geographical information.

(a) Plot the response, the number of military coups against each of the other variables.

(b) Use a stepwise AIC-based method to select a model that uses a smaller number of the available predictors.

(c) Does the deviance of your selected model indicate a good fit to the data?

(d) Make a QQ plot of the residuals and comment. Plot the residuals against the fitted values and interpret the result. What is the source of the lines of points observed on this plot?

(e) Give an interpretation of the coefficients of this plot.

(f) Count the number of countries with each number of military coups. Compare this with the numbers predicted by the previous model. Is there any evidence of excess of countries with zero coups? Use a Chi-squared test as implemented in `chisq.test()`.

5. The `dvisits` data comes from the Australian Health Survey of 1977–1978 and consist of 5190 single adults where young and old have been oversampled.

(a) Make plots which show the relationship between the response variable, `doctorco`, and the potential predictors, `age` and `illness`.

(b) Combine the predictors `chcond1` and `chcond2` into a single three-level factor. Make an appropriate plot showing the relationship between this factor and the response. Comment.

(c) Build a Poisson regression model with `doctorco` as the response and `sex`, `age`, `agesq`, `income`, `levyplus`, `freepoor`, `freerepa`, `illness`, `actdays`, `hscore` and the three-level condition factor as possible predictor variables. Considering the deviance of this model, does this model fit the data?

(d) Plot the residuals and the fitted values — why are there lines of observations on the plot? Make a QQ plot of the residuals and comment.

(e) Use a stepwise AIC-based model selection method. What sort of person would be predicted to visit the doctor the most under your selected model?

(f) For the last person in the dataset, compute the predicted probability distribution for their visits to the doctor, i.e., give the probability they visit 0, 1, 2, etc. times.

(g) Tabulate the frequencies of the number of doctor visits. Compute the expected frequencies of doctor visits under your most recent model. Compare the observed with the expected frequencies and comment on whether it is worth fitting a zero-inflated count model.

(h) Fit a comparable (Gaussian) linear model and graphically compare the fits. Describe how they differ.

6. Components are attached to an electronic circuit card assembly by a wave-soldering process. The soldering process involves baking and preheating the circuit card and then passing it through a solder wave by conveyor. Defects arise during the process. The design is 2^{7-3} with three replicates. The data is presented in the dataset `wavesolder`. You can assume that the replicates are independent.

(a) Make plots of the number of defects against each of the predictors. Comment on the relationships you see. Check graphically that there is no trend in the replicates.

(b) Compute the mean and variance within each group of three replicates. Plot the variance against the mean. Comment on the relationship and the viability of a Poisson model for the response. Repeat the plot, but use a log scale on both axes. Does this plot reveal anything new?

(c) Fit a Poisson model for the number of defects with all predictors included as main effects. What does the deviance of this model say about its fit?

(d) Make a plot of the residuals against the fitted values and comment on what is seen. Make a QQ plot of the residuals. Are there any outliers?

(e) Refit the Poisson model but excluding the case with the largest residual. Compute the deviance. Does this model now fit the data?

(f) Fit a quasi-poisson model with same model formula and excluded case. Estimate the value of the dispersion parameter. Check the model summary. Now use an F-test to the significance of each of the predictors. Compare the two sets of tests—one from the model summary and one from the F-test. Are they similar? Report on what predictors are significant and which level of the significant factors will lead to higher defects.

(g) Check the diagnostics again as in (d).

7. The dataset `esdcomp` was recorded on 44 doctors working in an emergency service at a hospital to study the factors affecting the number of complaints received.
   (a) Consider the rate of complaints in terms of the number received in relation to the number of patient visits made. Plot this against each of the four potential predictors and comment on the relationships.
   (b) Fit a binomial GLM for the number of complaints out of the number of visits with the other four variables as predictors. Does this model fit the data? Perform this check numerically and graphically.
   (c) Check the significance of the four predictors in the binomial model. Give a numerical interpretation to the effect of any significant predictors.
   (d) Fit an appropriate Poisson rate model for number of complaints that takes a comparable form to the binomial GLM fitted earlier. Does this model fit the data? Again check this numerically and graphically.
   (e) Again check the significance of the predictors and provide a numerical interpretation. Compare the conclusions of the two models.
   (f) Exchange the role of hours and visits in the Poisson rate model. Again check the significance of the predictors and interpret the outcome.
   (g) Compare the two proposed rate models.